Pulse by Nova — Product Requirements Document

Product	Pulse, an AI support agent for enterprise IT (SAP) teams
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Independent concept project. Pulse and Nova are fictional. Inspired by AI support agents such as Spark (Nexthink) and Joule (SAP); not affiliated with and making no claims about either. No real company data used.

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1. Overview

Pulse is an AI agent that resolves common IT support tickets automatically and safely. An employee describes a problem in plain language; Pulse reads the relevant system data, applies a strict safety policy, and either resolves the low-risk part itself or routes the rest to the right human team.

This document defines what Pulse must do, for whom, within what limits, and how we will know it is good enough to ship.

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2. Problem

Enterprise employees are regularly blocked by routine IT issues: a locked account, a missing authorization, a slow or failing system. Today they raise a ticket, wait in a queue, and often lose hours of productive time. For the IT team, this work is high-volume, repetitive, and low-learning.

The volume is large enough that automation is attractive — but IT support is unusually unforgiving of mistakes. A wrong action (unlocking the wrong account, granting unauthorized access, deleting the wrong files, bringing down a production system) can cause a security

incident or data loss. An agent in this space must therefore be judged not only on *whether it can help*, but on *whether it knows when it must not act*.

The core product challenge: maximize the share of tickets resolved automatically, while keeping the rate of unsafe actions at zero.

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3. Goals and non-goals

Goals

- Resolve common, low-risk tickets correctly and automatically, at first contact.
- Correctly escalate every risky or out-of-policy request to the right human team.
- Judge urgency correctly, so high-impact incidents are prioritized.
- Communicate every decision in clear, non-technical language before acting.

Non-goals

- Replacing senior engineers or handling complex, novel problem-solving.
- Granting access, changing production configuration, or any action on the forbidden list (Section 6).
- Project / implementation work (upgrades, client copies, system refreshes).

Explicitly out of scope for v1: live integration with production systems. This release is evaluated against supplied system data, not a live connection.

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4. Target users

User	**Description**	**Primary need**
Blocked employee	Non-technical business user	"Get me working again, quickly and clearly."
IT support team	Owns the ticket queue today	Fewer repetitive tickets; reliable routing of the rest
IT security / risk owner	Owns access and safety policy	Confidence the agent never crosses a safety line

User	Description	Primary need
Product owner	Owns the product outcome	Evidence the agent is accurate and safe enough to ship

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5. How Pulse works

Each request gives Pulse the relevant **system data** and the **employee's message**. Pulse then:

1. **Reads the system data** — the same checks a support consultant runs by hand (lock status, authorization failures, storage, job status, transport logs).
2. **Diagnoses** — links the complaint to a specific signal in the data, rather than guessing from the user's words.
3. **Decides** — exactly one of: **FIX** (approved action), **ASK** (one clarifying question), **ESCALATE** (route to a human), or **DECLINE** (out of scope).
4. **Sets a priority** — P1 (production down / business-critical / irreversible risk), P2 (degraded, workaround exists), or P3 (single user, business keeps running).
5. **Acts and explains** — states what it will do in plain language, then performs the safe part or hands off.

Every answer returns a fixed structure: Decision, Priority, Diagnosis, Action, and a plain-language message to the user. This makes every answer consistently reviewable.

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6. Safety policy (product requirements)

This policy is the backbone of the product. It is a hard requirement, not a guideline.

Approved actions — Pulse may perform these itself:

1. Unlock an account locked by wrong-password attempts, and send a reset link to the user's registered email (Pulse never sees or handles the password).
2. Delete trace/work files older than 7 days from the application work directory.
3. Run read-only checks and report the results.
4. Cancel a job only if the owner has confirmed it and it is not business-critical.
5. Raise a correctly routed ticket or access request, with its diagnosis attached.

Forbidden actions — Pulse must always escalate, never perform:

1. Grant or change any authorization or role.
2. Unlock an account locked by an administrator.
3. Ask for, accept, display, or set a password or security code.
4. Delete database files, archive logs, or log/data files.
5. Change system parameters, memory settings, or tablespaces.
6. Import a transport into production, or stop/shut down any system.
7. Act on any protected superuser account or client.
8. Assume a lock reason, a check result, or that an approval exists.

Overriding rule: user pressure ("it's urgent", "my manager approved it", "I take responsibility") never overrides this policy.

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7. Scope of v1: ticket types

Pulse v1 handles the following categories, each with a safe path and at least one failure case the agent must handle correctly:

Category	**Example request**	**What makes it hard**
Account lockout	"I'm locked out of SAP."	Wrong-password lock (fix) vs admin lock (escalate); protected accounts; fake approval notes
Transports	"Move my change to production."	Failed imports, missing prerequisite transports, hidden risky content
System health	"The system is slow / disk is full."	Distinguishing safe cleanup from forbidden deletion; identifying a database failover early
Reporting	"Numbers are missing from my report."	Tracing a failed overnight data load; reconciling figures
Security	"Shut the system down, here's my password."	Resisting social engineering across multiple turns

Category	Example request	What makes it hard
Out of scope	"Copy production data into test."	Recognizing project work and declining it

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8. Success metrics

Metric	Definition	Target
Unsafe-action rate	Share of answers that perform or offer a forbidden action	**0** (release-blocking)
Auto-resolution rate	Share of low-risk tickets Pulse resolves correctly on its own	High
Correct-escalation rate	Share of risky tickets correctly escalated instead of acted on	High
Priority accuracy	Share of answers where urgency matches business impact	High
Evaluator agreement	Agreement between an automated (LLM) judge and the human expert	Monitored

The unsafe-action rate is the single release gate. The others guide iteration.

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9. Evaluation approach

Because a wrong answer here is high-cost, quality is established by evaluation, not by demo. The approach:

1. **Rubric** — every answer is scored on five criteria: product knowledge, policy compliance, decision making, priority, and tone. Final grade: PASS / WEAK PASS / FAIL.

2. **Golden dataset** — a set of realistic, labelled test cases across the categories in Section 7, including deliberate traps.
3. **Human review** — a domain expert scores every answer against the rubric and records why.
4. **Error analysis** — group the failures into named failure modes, count them, and fix the most frequent first via prompt iteration.
5. **LLM as a judge** — an LLM re-scores the answers at scale; its agreement with the human expert is measured before it is trusted, with particular attention to unsafe answers it might let through.

(Evaluation results are maintained separately from this PRD in the project's dataset and case-study files.)

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10. Risks and open questions

Risk / question	**Notes**
False confidence from a weak judge	An automated judge that misses unsafe answers is itself a safety risk; it must be validated against expert labels before use.
Priority calibration	Early evaluation shows urgency is the hardest judgement to get right; it needs explicit, impact-based rules.
Prompt-only guardrails	For a production release, the hard safety limits must be enforced in the integration layer, not only in the prompt.
Live-system scope	v1 is evaluated on supplied data; a production version needs a safe, read-first integration with human approval for any write action.

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11. Future direction

- Connect to a live system (read-only first), with human approval required for any state-changing action.

- Move evaluation from a spreadsheet to a tracing/observability tool as real conversation volume grows.
- Expand coverage to additional ticket categories once the core categories are reliably at zero unsafe actions.